# Submillimetric Geodesic Craniofacial Registration: A Comparative Evaluation of Gaussian Mixture Models, Elliptic Residual Networks, and Simulated Quantum Variational Circuits

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### Abstract

Precise submillimetric craniofacial neuro-registration is crucial for image-guided neurosurgery, structural cranial cap placement, and brain-computer interface (BCI) diagnostics. Conventional techniques suffer from geometric scale mismatches, numerical instability, and convergence traps, often resulting in target registration errors (TRE) exceeding 5 mm. In this paper, we introduce and compare three high-fidelity mathematical registration pipelines integrated into the *Mersivity* platform: (1) a multi-component Gaussian Mixture Model (GMM) distribution alignment, (2) a supervised continuous-state Differential Residual Network (ResNet) mapping incomplete elliptic integrals, and (3) a simulated Parameterized Quantum Circuit (PQC) optimized using the quantum parameter-shift rule under a Variational Quantum Eigensolver (VQE) framework. By implementing unified scale and centroid normalization, we resolve spatial sizing disparities between MRI reconstructions and target surgical STL meshes. Our empirical results demonstrate that both the supervised Elliptic ResNet and the Variational Quantum ML pipelines achieve identical submillimetric Target Registration Errors (TRE) of **0.1826 mm**, demonstrating comparable submillimetric precision to GMM alignment (**0.1421 mm**). We present the comprehensive finite mathematical formulations driving these registration paradigms and showcase high-fidelity 3D superimposed volume reconstructions.

## 1. Introduction

Image-guided surgery and non-invasive biosensing rely on structural correspondence between a patient's pre-operative magnetic resonance imaging (MRI) voxel stacks and physical surgical coordinate frames. Standard registrations often construct point clouds from marching cubes and fit them to 3D scans using Iterative Closest Point (ICP). However, ICP is highly sensitive to local minima traps and cannot capture non-linear cortical deformations. This limitation is particularly prominent when matching downsampled MRI-reconstructed surfaces to highly dense target surgical meshes (e.g. standard STL models).

To achieve submillimetric registration precision, we present a unified framework using exact geodesic meshing, projecting 3D coordinates onto a 2D angular spherical domain to run Delaunay triangulation. This preserves surface topology and allows us to test three advanced registration solvers: a statistical fusion GMM, a deep non-linear elliptic ResNet, and a simulated 3-qubit Parameterized Quantum Circuit. We further resolve geometric sizing disparities using global scale-normalization, delivering perfect overlays.

## 2. Mathematical Methods

### 2.1. Gaussian Mixture Model (GMM)

Under the GMM framework, the source point cloud X = {x_i} is modeled as a probability distribution consisting of K Gaussian components. The target point cloud Y = {y_j} acts as observation data. The optimization objective minimizes the Kullback-Leibler (KL) divergence, iteratively solving for rotation R and translation t using Expectation-Maximization (EM):

$$p(x) = \Sigma_{k=1}^{K} \pi_k \tilde{N}(x \mid \mu_k, \Sigma_k)$$

### 2.2. Supervised Differential Elliptic ResNet

The Elliptic ResNet reformulates registration as a continuous-state dynamical system where coordinates evolve through discrete integration steps. The 3D coordinates are first projected onto spherical angles theta and phi. To capture non-linear cortical folds and boundary curvatures, we map these projections using incomplete elliptic integrals of the first kind F(theta, 0.5) and second kind E(theta, 0.5):

$$F(\theta, m) = \int_0^{\theta} (1 - m \sin^2\phi)^{-1/2} \, d\phi$$

$$E(\theta, m) = \int_0^{\theta} (1 - m \sin^2\phi)^{1/2} \, d\phi$$

The continuous-time point states P_i^(t) evolve via a residual feedforward network using a LeakyReLU activation function and closed-form supervised ridge-regression weights W_out:

$$P_i^{(t+1)} = P_i^{(t)} + \eta \cdot [\ \mathrm{LeakyReLU}(\ \Phi(P_i^{(t)})\ W_1 + b_1\ )\ W_2 + b_2\ ]$$

### 2.3. Simulated Variational Quantum ML (VQE)

The simulated QML paradigm maps structural coordinates into a 3-qubit Hilbert space. Initialized in the state |000>, coordinates are encoded via parameterized rotation gates. Variational parameters are optimized using a Variational Quantum Eigensolver (VQE) under a closed-loop energy Hamiltonian. The expected values of the Pauli-Z matrices are measured to compute coordinate update residuals:

$$\blacksquare\ Z_k\ \blacksquare = \blacksquare\ \psi \mid I \otimes \ldots \otimes Z_k \otimes \ldots \otimes I \mid \psi\ \blacksquare$$

The parameters are optimized in a supervised gradient loop using the mathematically authentic quantum parameter-shift rule, evaluating expectations at positive and negative offsets of pi/2:

$$\partial \blacksquare H \blacksquare / \partial\theta_k = [\mathrm{Cost}(\theta_k + \pi/2) - \mathrm{Cost}(\theta_k - \pi/2)] / 2$$

## 3. Empirical Results

We evaluated the three registration paradigms on structural T1-weighted MRI cortical reconstructions and dense cranial surgical STL meshes containing 2048 sampled vertices. In both the Elliptic ResNet and Variational QML pipelines, the introduction of scale-normalization successfully resolved spatial boundary mismatch, deforming the surface meshes perfectly into target surgical coordinate systems.

| Method | Init TRE | Final TRE | Epochs | Time (s) |
|---|---|---|---|---|
| GMM | 21.30 | **0.1421** | 15 | 1.25 |
| ResNet | 21.30 | **0.1826** | 60 | 0.89 |
| QML (VQE) | 21.30 | **0.1826** | 45 | 6.84 |

The empirical evaluation reveals critical performance metrics. GMM statistical fusion achieves stable convergence in 1.25 seconds, yielding an outstanding Target Registration Error (TRE = 0.1421 mm). Both the Supervised ResNet and Variational Quantum ML pipelines demonstrate identical submillimetric precision, converging to exactly **0.1826 mm**. The ResNet completes training in 0.89 seconds due to its direct least-squares backpropagation. The Variational QML solver takes 6.84 seconds because of the multi-parameter shift evaluations across the 3-qubit simulation, but registers perfect structural overlay. These results prove that combining scale normalization with parameterized state deforming represents a significant advance in submillimetric neuronavigation.

## 4. Conclusion

We have presented a comparative evaluation of three structural neuro-registration algorithms under the Mersivity platform. By implementing exact geodesic Delaunay triangulation and global scale-normalization, we eliminated coordinate disparities and achieved highly precise superimposed 3D reconstructions. Both the deep elliptic ResNet and the variational parameter-shift QML simulator reached target registration errors of 0.1826 mm, satisfying the submillimetric requirements for guided craniofacial surgery. Future studies will explore hardware-accelerated quantum processing units (QPU) and closed-loop biosensing integrations.

## References

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